

Notes on Gauge Theory

Bob

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1 Affine Connections

1.1 Levi-Civita Connections

When it comes to connections, most of us probably think first of the Levi-Civita connection in Riemannian geometry. So let us first review its most basic definition, and then try to rewrite everything we are familiar with in the language of differential forms.

1.1.1 Covariant Derivative

Definition 1.1. A connection ∇ on a manifold M is a map

$$\begin{aligned}\nabla : C^\infty(M, TM) \times C^\infty(M, TM) &\rightarrow C^\infty(M, TM) \\ (Y, X) &\mapsto \nabla_Y X\end{aligned}$$

satisfying the following properties:

$$\begin{aligned}\nabla_{f_1 Y_1 + f_2 Y_2} X &= f_1 \nabla_{Y_1} X + f_2 \nabla_{Y_2} X, \\ \nabla_Y (\lambda_1 X_1 + \lambda_2 X_2) &= \lambda_1 \nabla_Y X_1 + \lambda_2 \nabla_Y X_2, \\ \nabla_Y f X &= df(Y) \cdot X + f \nabla_Y X.\end{aligned}$$

where $X, X_1, X_2, Y, Y_1, Y_2 \in \Gamma(TM)$, $f, f_1, f_2 \in C^\infty(M)$, and $\lambda_1, \lambda_2 \in \mathbb{R}$.

Definition 1.2. The **torsion**¹ of a connection ∇ is defined as the tensor

$$T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y].$$

The **curvature** of a connection ∇ is defined as the tensor

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Definition 1.3. A connection ∇ of a Riemannian manifold (M, g) is called a *Levi-Civita connection* if it is torsion-free and compatible with the metric g , i.e.

$$\begin{aligned}\nabla_X Y - \nabla_Y X &= [X, Y], \\ X(g(Y, Z)) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z).\end{aligned}$$

¹It can only be defined for connections on the tangent bundle, since it involves both $\nabla_X Y$ and $\nabla_Y X$.

1.1.2 Exterior covariant derivative

A connection ∇ can also be viewed as a map

$$\begin{aligned}\nabla : C^\infty(M, TM) &\rightarrow C^\infty(M, T^*M \otimes TM) \\ X &\mapsto \nabla X\end{aligned}$$

where ∇X is a 1-form taken value in TM^1 and $(\nabla X)(Y) := \nabla_Y X$. ∇ is not a $C^\infty(M)$ -linear map, but

$$\nabla(fX) = df \otimes X + f\nabla X.$$

We can extend ∇ to the exterior covariant derivative

$$\begin{aligned}\nabla : \Omega^*(M, TM) &\rightarrow \Omega^{*+1}(M, TM)^2 \\ \eta &\mapsto \nabla \eta,\end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta, \quad \forall \alpha \in \Omega^q(M), \eta \in \Omega^*(M, TM),$$

For example, let $\eta = \alpha \otimes X$ be a section of $\Omega^r(M, TM)$, then

$$\nabla \eta = \nabla(\alpha \otimes X) = d\alpha \otimes X + (-1)^r \alpha \wedge \nabla X.$$

Under the above setup, we can define $\nabla^2 = \nabla \circ \nabla, \nabla^3, \dots, \nabla^k$, and we can prove the following property:

Proposition 1.4. $\nabla^2 : \Omega^*(M, TM) \rightarrow \Omega^{*+2}(M, TM)$ is an $\Omega^*(M)$ -linear map, i.e.

$$\nabla^2(\alpha \wedge \eta) = \alpha \wedge \nabla^2 \eta, \quad \forall \alpha \in \Omega^*(M), \eta \in \Omega^*(M, TM).$$

Thus $R := \nabla^2$ defines a section of $\Omega^2(M, \text{End}(TM))$, which is called the **curvature** of the connection ∇ .

Proof.

$$\begin{aligned}\nabla^2(\alpha \wedge \eta) &= \nabla(d\alpha \wedge \eta + (-1)^q \alpha \wedge d^\nabla \eta) \\ &= d^2 \alpha \wedge \eta + (-1)^{q+1} d\alpha \wedge d^\nabla \eta + (-1)^q d\alpha \wedge \nabla \eta + \alpha \wedge \nabla^2 \eta \\ &= \alpha \wedge \nabla^2 \eta.\end{aligned}$$

□

¹One can simply view ∇X as a $(1, 1)$ -tensor field.

² $\Omega^r(M, TM) := C^\infty(M, \wedge^r T^*M \otimes TM)$.

Unlike the usual exterior derivative d , we have $\nabla^2 \neq 0$ in general. The reason why ∇^2 is called the curvature of ∇ is that

Proposition 1.5. *For $X, Y, Z \in C^\infty(M, TM)$, we have*

$$(\nabla^2 Z)(X, Y) = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

Proof. We will prove it after introducing the connection 1-forms. $\square$

1.1.3 Connection 1-form

Locally, we always have a local frame. Let $\{e_i\}$ be a local frame of TM on an open set $U \subset M$, and $\{\omega^i\}$ be the dual frame. then we can write

$$\nabla_{e_i} e_j = \Gamma_{ij}^k e_k,$$

where Γ_{ij}^k are the Christoffel symbols of the connection ∇ w.r.t. $\{e_i\}$. Define

$$\omega_j^k := \Gamma_{ij}^k \omega^i,$$

we call ω_j^k the **connection 1-form** of ∇ w.r.t. $\{e_i\}$. So

$$\nabla e_j = \omega_j^k \otimes e_k.$$

Let the matrix of 1-forms $\omega = (\omega_j^k)^1$, then we can write the above equation in a more compact way:

$$\nabla(e_1, \dots, e_n) = (e_1, \dots, e_n) \omega = (e_1, \dots, e_n) \begin{pmatrix} \omega_1^1 & \cdots & \omega_1^n \\ \vdots & & \vdots \\ \omega_n^1 & \cdots & \omega_n^n \end{pmatrix}$$

The connection 1-form ω is not a tensor, since it depends on the choice of the local frame. Let $\{\tilde{e}_j\}$ be another local frame, and the transition matrix from $\{e_i\}$ to $\{\tilde{e}_j\}$ is $a = (a_i^j)$, i.e.

$$(\tilde{e}_1, \dots, \tilde{e}_n) = (e_1, \dots, e_n) a,$$

then

$$\begin{aligned} (\tilde{e}_1, \dots, \tilde{e}_n) \tilde{\omega} &= \nabla(\tilde{e}_1, \dots, \tilde{e}_n) = \nabla((e_1, \dots, e_n) a) \\ &= (\nabla(e_1, \dots, e_n)) a + (e_1, \dots, e_n) da \\ &= (e_1, \dots, e_n) \omega a + (e_1, \dots, e_n) da \end{aligned}$$

¹The upper index k is the row index, and the lower index j is the column index.

$$=(\tilde{e}_1, \dots, \tilde{e}_n)(a^{-1}\omega a + da).$$

Thus

$$\tilde{\omega} = a^{-1}da + a^{-1}\omega a. \quad (1)$$

For $\{e_i\} = \{\frac{\partial}{\partial x^i}\}$, $\{\tilde{e}_j\} = \{\frac{\partial}{\partial \tilde{x}^j}\}$, $\omega_j^k = \Gamma_{ij}^k dx^i$, the above formula reduces to the well-known transformation formula of Christoffel symbols:

$$\tilde{\Gamma}_{ij}^k = \frac{\partial \tilde{x}^k}{\partial x^l} \frac{\partial x^m}{\partial \tilde{x}^i} \frac{\partial x^n}{\partial \tilde{x}^j} \Gamma_{mn}^l + \frac{\partial^2 x^l}{\partial \tilde{x}^i \partial \tilde{x}^j} \frac{\partial \tilde{x}^k}{\partial x^l}.$$

We can compute the curvature in terms of the connection 1-form ω :

$$\begin{aligned} \nabla^2(e_1, \dots, e_n) &= \nabla(\nabla(e_1, \dots, e_n)) = \nabla((e_1, \dots, e_n)\omega) \\ &= (\nabla(e_1, \dots, e_n)) \wedge \omega + (e_1, \dots, e_n) d\omega \\ &= (e_1, \dots, e_n)\omega \wedge \omega + (e_1, \dots, e_n) d\omega \\ &= (e_1, \dots, e_n)(d\omega + \omega \wedge \omega). \end{aligned}$$

Define

$$\Omega := d\omega + \omega \wedge \omega, \quad (2)$$

then Ω is a matrix of 2-forms, and we call Ω the **curvature 2-form** of the connection ∇ w.r.t. $\{e_i\}$.

Proposition 1.6. *Let $R(e_i, e_j)e_k = R_{ijk}^l e_l$, then*

$$\Omega_k^l(e_i, e_j) = R_{ijk}^l.$$

Proof. Since $\Omega_k^l = d\omega_k^l + \omega_m^l \wedge \omega_k^m$,

$$\begin{aligned} (d\omega_k^l)(e_i, e_j)e_l &= e_i(\omega_k^l(e_j))e_l - e_j(\omega_k^l(e_i))e_l - \omega_k^l([e_i, e_j])e_l \\ &= \nabla_{e_i}(\omega_k^l(e_j)e_l) - \omega_k^l(e_j)\nabla_{e_i}e_l - \nabla_{e_j}(\omega_k^l(e_i)e_l) + \omega_k^l(e_i)\nabla_{e_j}e_l - \omega_k^l([e_i, e_j])e_l \\ &= \nabla_{e_i}\nabla_{e_j}e_k - \omega_k^l(e_j)\nabla_{e_i}e_l - \nabla_{e_j}\nabla_{e_i}e_k + \omega_k^l(e_i)\nabla_{e_j}e_l - \omega_k^l([e_i, e_j])e_l, \end{aligned}$$

and

$$\begin{aligned} (\omega_m^l \wedge \omega_k^m)(e_i, e_j)e_l &= \omega_m^l(e_i)\omega_k^m(e_j)e_l - \omega_m^l(e_j)\omega_k^m(e_i)e_l \\ &= \omega_k^m(e_j)\nabla_{e_i}e_m - \omega_k^m(e_i)\nabla_{e_j}e_m. \end{aligned}$$

Thus

$$\Omega_k^l(e_i, e_j)e_l = \nabla_{e_i}\nabla_{e_j}e_k - \nabla_{e_j}\nabla_{e_i}e_k - \nabla_{[e_i, e_j]}e_k = R(e_i, e_j)e_k.$$

□

Corollary 1.7. *This proposition immediately implies Prop 1.5 since*

$$(\nabla^2 e_k)(e_i, e_j) = \Omega_k^l(e_i, e_j)e_l = R(e_i, e_j)e_k.$$

Applying the transformation formula (1) to (2), we get

$$\begin{aligned}\tilde{\Omega} &= d\tilde{\omega} + \tilde{\omega} \wedge \tilde{\omega} \\ &= d(a^{-1}da + a^{-1}\omega a) + (a^{-1}da + a^{-1}\omega a) \wedge (a^{-1}da + a^{-1}\omega a) \\ &= da^{-1} \wedge da + da^{-1} \wedge \omega a + a^{-1}d\omega a - a^{-1}\omega \wedge da \\ &\quad + a^{-1}da \wedge a^{-1}da + a^{-1}da \wedge a^{-1}\omega a + a^{-1}\omega \wedge da + a^{-1}\omega \wedge \omega a,\end{aligned}$$

since

$$a^{-1}a = I \implies da^{-1}a + a^{-1}da = 0 \implies da^{-1} = -a^{-1}da a^{-1},$$

we have

$$\begin{aligned}\tilde{\Omega} &= -a^{-1}da a^{-1} \wedge da - a^{-1}da a^{-1} \wedge \omega a + a^{-1}d\omega a - a^{-1}\omega \wedge da \\ &\quad + a^{-1}da \wedge a^{-1}da + a^{-1}da \wedge a^{-1}\omega a + a^{-1}\omega \wedge da + a^{-1}\omega \wedge \omega a \\ &= a^{-1}(d\omega + \omega \wedge \omega)a = a^{-1}\Omega a.\end{aligned}$$

Thus transformation formula for curvature between $\{e_i\}$ and $\{\tilde{e}_j\}$ is given by

$$\tilde{\Omega} = a^{-1}\Omega a. \quad (3)$$

The above formula shows that although Ω isn't a globally defined form, two different curvature forms only differ by a conjugation. So if we apply a quantity invariant under conjugation, we obtain a globally defined form. This is the idea of Chern-Weil theory, which we will discuss in later section.

For a Riemannian manifold (M, g) , let ∇ be the Levi-Civita connection. We can take $\{e_i\}$ to be a local orthonormal frame, and $\{\omega^i\}$ be the corresponding dual frame. Then

$$\begin{aligned}\langle \nabla_{e_k} e_i, e_j \rangle + \langle e_i, \nabla_{e_k} e_j \rangle &= e_k \langle e_i, e_j \rangle = 0 \\ \implies \langle \omega_i^l(e_k) e_l, e_j \rangle + \langle e_i, \omega_j^l(e_k) e_l \rangle &= 0 \\ \implies \omega_i^j(e_k) + \omega_j^i(e_k) &= 0.\end{aligned}$$

So the connection 1-form ω w.r.t. an orthonormal frame is skew-symmetric, i.e. $\omega \in \Omega^1(M, \mathfrak{so}(n))$. Since $\Omega = d\omega + \omega \wedge \omega$, the curvature 2-form Ω is also skew-symmetric, i.e. $\Omega \in \Omega^2(M, \mathfrak{so}(n))$.

1.2 General Affine Connections

Let $\pi : E \rightarrow M$ be a vector bundle, we can also define affine connections on E in a similar way as we did for the tangent bundle.

1.2.1 Covariant Derivative

The content of this section is almost a verbatim copy of the previous content.

Definition 1.8. An (affine) connection ∇ on a vector bundle E is a map

$$\begin{aligned} \nabla : C^\infty(M, TM) \times C^\infty(M, E) &\rightarrow C^\infty(M, E) \\ (X, s) &\mapsto \nabla_X s \end{aligned}$$

satisfying

$$\begin{aligned} \nabla_{f_1 X_1 + f_2 X_2} s &= f_1 \nabla_{X_1} s + f_2 \nabla_{X_2} s, \\ \nabla_X (\lambda_1 s_1 + \lambda_2 s_2) &= \lambda_1 \nabla_X s_1 + \lambda_2 \nabla_X s_2, \\ \nabla_X (fs) &= df(X) \cdot s + f \nabla_X s, \end{aligned}$$

The existence of connections on any vector bundle can be proved by using partitions of unity, which we left as an exercise.

The reason why we call ∇ an affine connection is that given two connections ∇_1 and ∇_2 on E , $(1-t)\nabla_1 + t\nabla_2$ is also a connection on E for any $t \in \mathbb{R}$.

Let $(\nabla s)(X) := \nabla_X s$, we can extend ∇ to the exterior covariant derivative

$$\begin{aligned} \nabla : \Omega^*(M, E) &\rightarrow \Omega^{*+1}(M, E) \\ \eta &\mapsto \nabla \eta, \end{aligned}$$

satisfying the Leibniz rule:

$$\nabla(\alpha \wedge \eta) = d\alpha \wedge \eta + (-1)^q \alpha \wedge \nabla \eta, \quad \forall \alpha \in \Omega^q(M), \eta \in \Omega^*(M, E).$$

1.2.2 Connection 1-form

1.2.3 Horizontal Distribution

1.2.4 Curvature of Connection

Exercises

Exercise 1.1. In Riemannian geometry, we can define the covariant derivative of a 1-form ω by

$$(\nabla_X \omega)(Y) := X(\omega(Y)) - \omega(\nabla_X Y).$$

For an (r, s) -tensor field T , the covariant derivative $\nabla_Z T$ is

$$\begin{aligned} (\nabla_Z T)(X_1, \dots, X_r, \omega^1, \dots, \omega^s) &:= Z(T(X_1, \dots, X_r, \omega^1, \dots, \omega^s)) \\ &\quad - \sum_{i=1}^r T(X_1, \dots, \nabla_Z X_i, \dots, X_r, \omega^1, \dots, \omega^s) \\ &\quad - \sum_{j=1}^s T(X_1, \dots, X_r, \omega^1, \dots, \nabla_Z \omega^j, \dots, \omega^s). \end{aligned}$$

If we rearrange the terms, we see that this is precisely the Leibniz rule.

The curvature operator can also act on general tensors

$$R(X, Y)T := \nabla_X \nabla_Y T - \nabla_Y \nabla_X T - \nabla_{[X, Y]} T.$$

We can define the covariant differential $\nabla_{\text{tensor}} T$ as the $(r, s + 1)$ -tensor field

$$(\nabla_{\text{tensor}} T)(Z, \dots, X_i, \dots, \dots, \omega^j, \dots) := (\nabla_Z T)(\dots, X_i, \dots, \dots, \omega^j, \dots)$$

This is another way to extend the covariant derivative to tensor fields, and we denote it by ∇_{tensor} to distinguish it from the exterior covariant derivative.

Prove that for a torsion-free connection, ∇_{tensor} satisfies the Ricci identity

$$(\nabla_{\text{tensor}}^2 T)(X, Y, \dots, \dots) - (\nabla_{\text{tensor}}^2 T)(Y, X, \dots, \dots) = (R(X, Y)T)(\dots, \dots)$$

Proof. For an (r, s) -tensor field T , we have

$$\begin{aligned} &(\nabla_{\text{tensor}}^2 T)(X, Y, \dots, \dots) = (\nabla_X (\nabla_{\text{tensor}} T))(Y, \dots, \dots) \\ &= X((\nabla_{\text{tensor}} T)(Y, \dots, \dots)) - (\nabla_{\text{tensor}} T)(\nabla_X Y, \dots, \dots) \\ &\quad - \sum_{i=1}^r (\nabla_{\text{tensor}} T)(Y, \dots, \nabla_X X_i, \dots, \dots) - \sum_{j=1}^s (\nabla_{\text{tensor}} T)(Y, \dots, \dots, \nabla_X \omega^j, \dots) \\ &= X((\nabla_Y T)(\dots, \dots)) - (\nabla_{\nabla_X Y} T)(\dots, \dots) \\ &\quad - \sum_{i=1}^r (\nabla_Y T)(\dots, \nabla_X X_i, \dots, \dots) - \sum_{j=1}^s (\nabla_Y T)(\dots, \dots, \nabla_X \omega^j, \dots) \\ &= (\nabla_X \nabla_Y T)(\dots, \dots) - (\nabla_{\nabla_X Y} T)(\dots, \dots), \end{aligned}$$

Thus

$$\begin{aligned} &(\nabla_{\text{tensor}}^2 T)(X, Y, \dots, \dots) - (\nabla_{\text{tensor}}^2 T)(Y, X, \dots, \dots) \\ &= (\nabla_X \nabla_Y T)(\dots, \dots) - (\nabla_Y \nabla_X T)(\dots, \dots) - (\nabla_{[X, Y]} T)(\dots, \dots) \\ &= (R(X, Y)T)(\dots, \dots). \end{aligned}$$

□

Exercise 1.2. For a vector field $Z \in C^\infty(M, TM)$, both $\nabla^2 Z$ and $\nabla_{\text{tensor}}^2 Z$ are $(1, 2)$ -tensor fields, but by Ricci identity, we have

$$(\nabla_{\text{tensor}}^2 Z)(X, Y) - (\nabla_{\text{tensor}}^2 Z)(Y, X) = R(X, Y)Z = (\nabla^2 Z)(X, Y).$$

Can you explain this phenomenon?